Reg. No. | | | | | | | | | | |

[18ESEE02]

FIRST SEMESTER B.Tech EXAMINATION, DECEMBER–2018

BASICS OF ELECTRICAL ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions:
i. Answer all the Questions (Q.No1 & Q.No 2 are compulsory)
ii. Missing data may be suitably assumed.

1. a. State and Explain Kirchhoff's laws. **05**

   b. A resistance R is connected in series with a parallel circuit comprising of two resistances of 14 ohms & 8 ohms. The total power dissipated in the circuit is 70 Watts when the applied voltage is 24 volts, calculate R. **04**

   c. Two batteries of 12V and 12.5V with internal resistance of 0.1 ohm and 0.2 ohm are connected in parallel. When a resistance of 1 ohm is connected across the above circuit. Find the current through it. **05**

2. a. Prove that a pure capacitor does not consume any real power, when connected across the A C voltage, with relevant wave forms. **06**

   b. Three equal impedances having resistance of 6 ohms and inductive reactance of 4 ohms are connected in delta across three phase 440V supply. Find **08**
      i. Phase current
      ii. Line current
      iii. Power consumed in each phase.

3. a. Explain the principle of operation of a transformer. With neat sketch explain the constructional features of different types of transformer. **06**

   b. What is the difference between ideal transformer and practical transformer? **04**

OR

4. a. What is a transformer? With neat sketch, explain the construction features of transformer. **04**

   b. State the different types of losses in the transformer, also how to minimize them. **04**

   c. With a neat diagram explain the equivalent circuit of transformer. **06**

5. a. With neat sketch explain the construction and working of a three phase induction motor. **06**

   b. Explain the necessity of starter. With neat sketch explain Star-Delta starter. **04**

   c. Write the comparison between squirrel cage and phase wound Induction motor. **04**

OR

6. a. What are the advantages of rotating field of synchronous generator? **04**

b. List the differences between salient and non-salient type rotors. **05**

c. With circuit diagram, explain the speed control of separately excited DC motor. **05**

7. a. With neat circuit diagrams, explain two-way and three-way control of a lamp. **08**

b. What is Earthing? State some important points on the necessity of Earthing. **06**

OR

8. a. Briefly explain: **08**

i. Switch fuse unit

ii. MCB,

iii. ELCB,

iv. MCCB.

b. Define Power factor? State the significance of power factor. List out the methods of improve p.f. **06**